THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of time is the: 1

(a) minute **(b)** second **(c)** day **(d)** hour

2. Speed of an object is: 1

(a) total time **(b)** total distance

(c) distance travelled per unit time **(d)** time per unit distance

3. The instrument in a vehicle which records the distance travelled is the: 1

(a) barometer **(b)** speedometer **(c)** thermometer **(d)** odometer

4. 1 hour is equal to: 1

(a) 600 seconds **(b)** 60 seconds **(c)** 360 seconds **(d)** 3600 seconds

5. The SI unit of speed is: 1

(a) kilogram **(b)** kilometre **(c)** metre per second **(d)** second

6. The instrument in a vehicle which shows the speed is the: 1

(a) compass **(b)** clock **(c)** odometer **(d)** speedometer

7. The time taken to complete one oscillation of a pendulum is its: 1

(a) amplitude **(b)** time period **(c)** frequency **(d)** speed

8. Another common unit of speed is: 1

(a) metre **(b)** kilometre

(c) kilometre per hour **(d)** hour

9. A graph showing distance covered at different times is a: 1

(a) bar chart **(b)** histogram

(c) distance-time graph **(d)** pie chart

10. One complete to-and-fro motion of a pendulum bob is called an: 1

(a) revolution **(b)** vibration of sound only

(c) oscillation **(d)** rotation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define speed and write its SI unit. 2

12. A boy runs 400 m in 80 s. Find his speed. 2

13. Differentiate between uniform and non-uniform motion. 2

14. Define time period of a pendulum. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A pendulum completes 30 oscillations in 60 s. Find its time period and the number of oscillations in 1 minute. 3

16. Convert: 72 km/h to m/s and 15 m/s to km/h. 3

17. A cyclist rides at 18 km/h. How far does he go in 20 minutes? 3

OR

A bus covers 180 km in 4 h. Find its speed in km/h and m/s. 3

18. A train takes 5 h to cover 400 km. Find its average speed and the distance in 3 h at that speed. 3

OR

Explain how the speedometer and odometer help the driver. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the experiment to find the time period of a simple pendulum and show how it depends on length. 5

OR

Describe the oscillatory motion of a pendulum and explain terms: oscillation, time period, amplitude. 5

20. A car travels at 60 km/h for 2 h, then at 40 km/h for 3 h. Find the total distance, total time and average speed. 5

OR

Convert and compare the speeds: a snail 0.5 cm/s, a cyclist 18 km/h, a car 90 km/h. 5

END OF QUESTION PAPER — Total: 40 Marks

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